

Public Transport Network Analysis

Graph Structure, Temporal Dynamics and Accessibility

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1 Project Overview

This project studies public transport systems as graph-based networks. Four cities are considered: **Canberra, Detroit, Paris and Rome**. For each city, stops are represented as nodes and direct transport connections as directed edges. In addition to the static graph, temporal information is retained for every connection, including departure time, arrival time and travel duration. This makes it possible to combine classical graph analysis with time-dependent accessibility analysis.

Two datasets are used for the temporal network: a `day.csv` file describing one day of service and a `week.csv` file describing a full week. The project addresses five main tasks: identifying the busiest departure stops, finding nodes with the highest out-degree, computing the graph diameter, analysing the number of connections over time and evaluating how many stops can be reached within fixed time windows.

2 Data Representation and Graph Construction

For each city, the input files contain a list of stops and a list of temporal connections. A directed graph $G = (V, E)$ is created, where V is the set of public transport stops and E contains the direct connections between them. Additional dictionaries store the stop names, geographic coordinates and temporal edge information. The temporal edge structure can be represented as tuples of the form

$$(u, v, t_{\text{dep}}, t_{\text{arr}}, \Delta t),$$

where u and v are the origin and destination stops, t_{dep} and t_{arr} are departure and arrival times, and Δt is the connection duration.

This design separates the network topology from the time-dependent information while keeping both readily available for later algorithms.

3 Network Visualisation

The geographic layouts reveal substantial differences in network structure. Canberra appears relatively sparse and spatially fragmented, while Paris and Rome contain much denser central regions. Detroit displays a more regular, grid-like structure in several parts of the network.

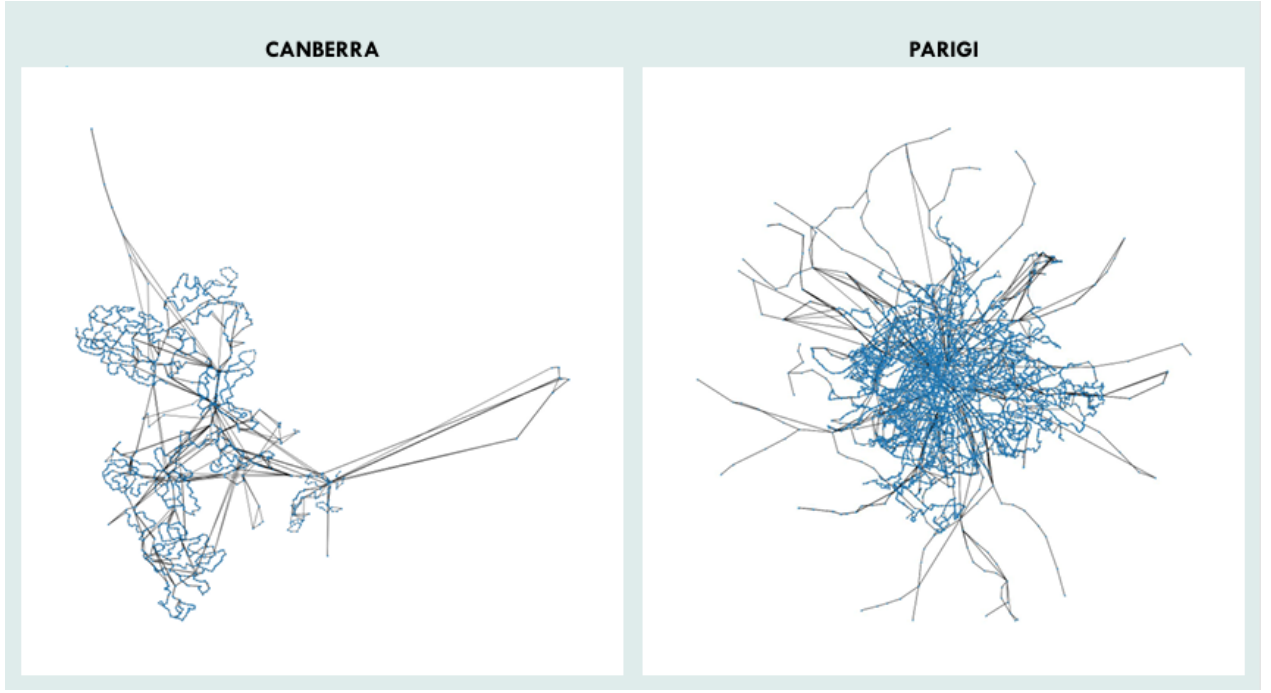


Figure 1: Public transport network graphs for Canberra and Paris.

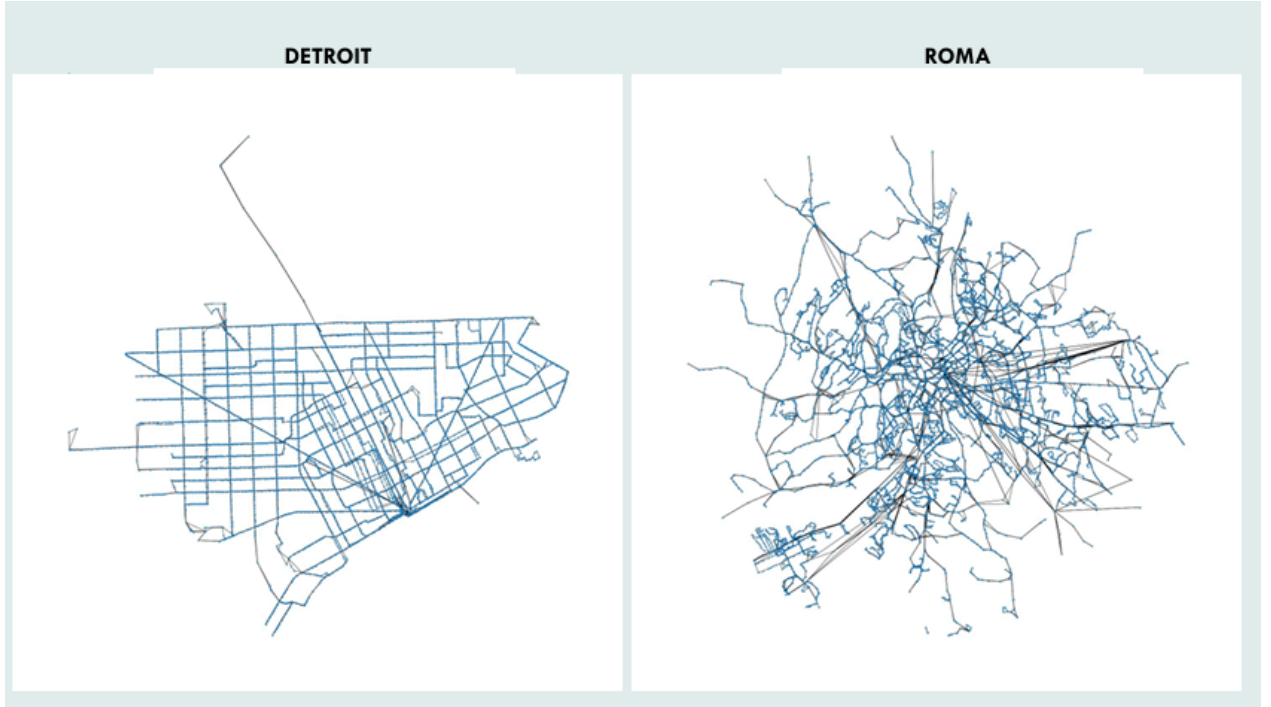


Figure 2: Public transport network graphs for Detroit and Rome.

4 Task 1: Stops with the Largest Number of Departures

The first analysis counts how many trips depart from each stop. For every directed edge, the counter associated with its origin stop is increased. The ten largest values are then selected. The procedure scans the edge list once and then performs a small fixed-size top-10 selection, giving an overall complexity of approximately $O(C(n + m))$, where C is the number of cities, n the number of nodes and m the number of edges.

The Canberra results show that the ranking is highly stable between the daily and weekly datasets. Weekly departure counts are approximately 5.4–5.9 times the daily values for the leading stops.

Table 1: Top departure stops in Canberra.

Stop	Week	Day	Week/Day
Callum St after Hindmarsh Dr	2963	524	5.654
Belc. Community Stn Arrivals	2950	502	5.877
Westfield Bus Stn Set Down Only	2950	502	5.877
Westfield Bus Stn	2904	508	5.717
London Cct Legislative Assembly	2845	508	5.599
Eastern Valley Way before Emu Bank Dr	2785	480	5.802
London Cct Commonwealth Bank	2632	474	5.553
Alinga St Health Building (Plt 13)	2522	464	5.433
Launceston St opp Phillip Oval	2462	430	5.726
Eastern Valley Way after Emu Bank Dr	2168	382	5.678

5 Task 2: Nodes with the Highest Out-Degree

Out-degree measures the number of distinct directed edges leaving a node. This differs from the number of departures: a stop can have many scheduled trips but only a limited number of distinct destinations. The algorithm computes `G.out_degree()` for all nodes and iteratively selects the ten highest values.

In Canberra, *Civic Interchange Bay 6*, *Alinga St* has the largest out-degree, equal to 8 in both datasets. Several other central stops have values between 5 and 7, indicating that they act as important branching points in the network.

Table 2: Canberra nodes with the highest out-degree.

Node	ID	Day degree	Week degree
Civic Interchange Bay 6, Alinga St	14026	8	8
London Cct Legislative Assembly	1486	7	7
Morisset St After Crawford St	32495	7	7
Canberra Centre City	1249	6	6
Soward Way Homeworld	57	5	5
Comrie St opp Erindale Centre Plt 2	161	5	7
Launceston St Phillip Oval	651	5	5
Brierly St Cooleman Court Plt 2	1103	5	5
Corinna St Westfield Woden	1160	5	—
Streeton Drive after Unwin Pl	1196	5	—

6 Task 3: Graph Diameter

The graph diameter is the maximum shortest-path distance between two nodes in the main connected component. Because the transport graphs are directed, they are first converted to undirected graphs for this analysis. The largest connected component is extracted, and a two-sweep breadth-first search provides an initial estimate and a central starting point. The iFUB procedure then refines lower and upper bounds through targeted BFS explorations until the exact diameter is obtained.

Table 3: Diameter of the largest connected component in the daily network.

City	Diameter	Nodes	Starting stop
Canberra	51	2520	Northbourne Av before Macarthur Av (2043)
Detroit	206	5683	ROSA PARKS & SPRUCE (5305)
Paris	159	10644	LA BOULE BLANCHE (4767)
Rome	99	7562	CAVALLEGGERI/S. PIETRO (5992)

The values show that network size and diameter are not equivalent concepts. Paris has the largest main component, but Detroit has the largest diameter, indicating longer shortest-path chains across its connected structure.

7 Task 4: Temporal Distribution of Connections

The fourth task analyses how service activity changes over time. Departure timestamps are converted to datetime values, grouped into one-hour intervals and counted. The same procedure is applied to both the daily and weekly datasets. The resulting profiles highlight recurring daily peaks and the reduction in service during lower-demand periods.

7.1 Canberra

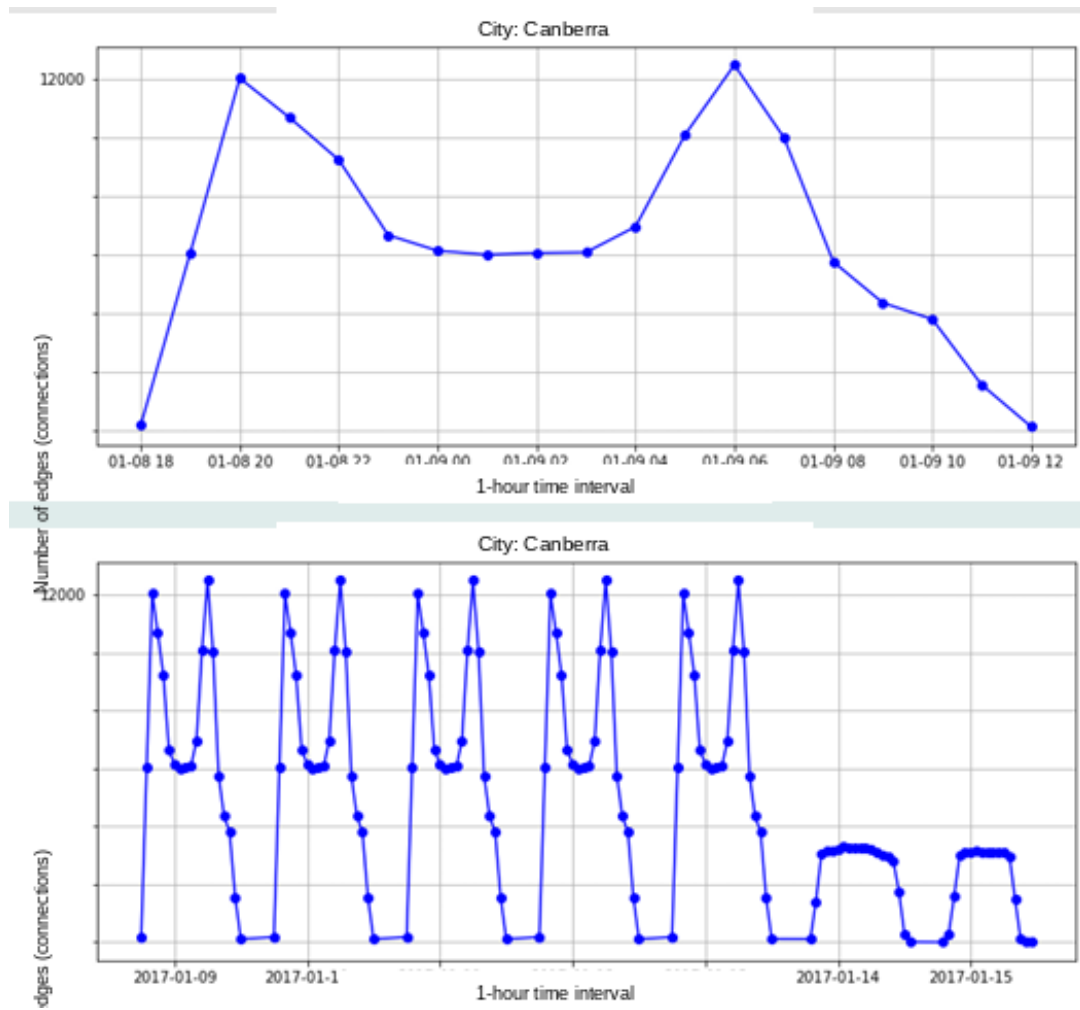


Figure 3: Number of connections by one-hour interval in Canberra: daily file (top) and weekly file (bottom).

7.2 Detroit

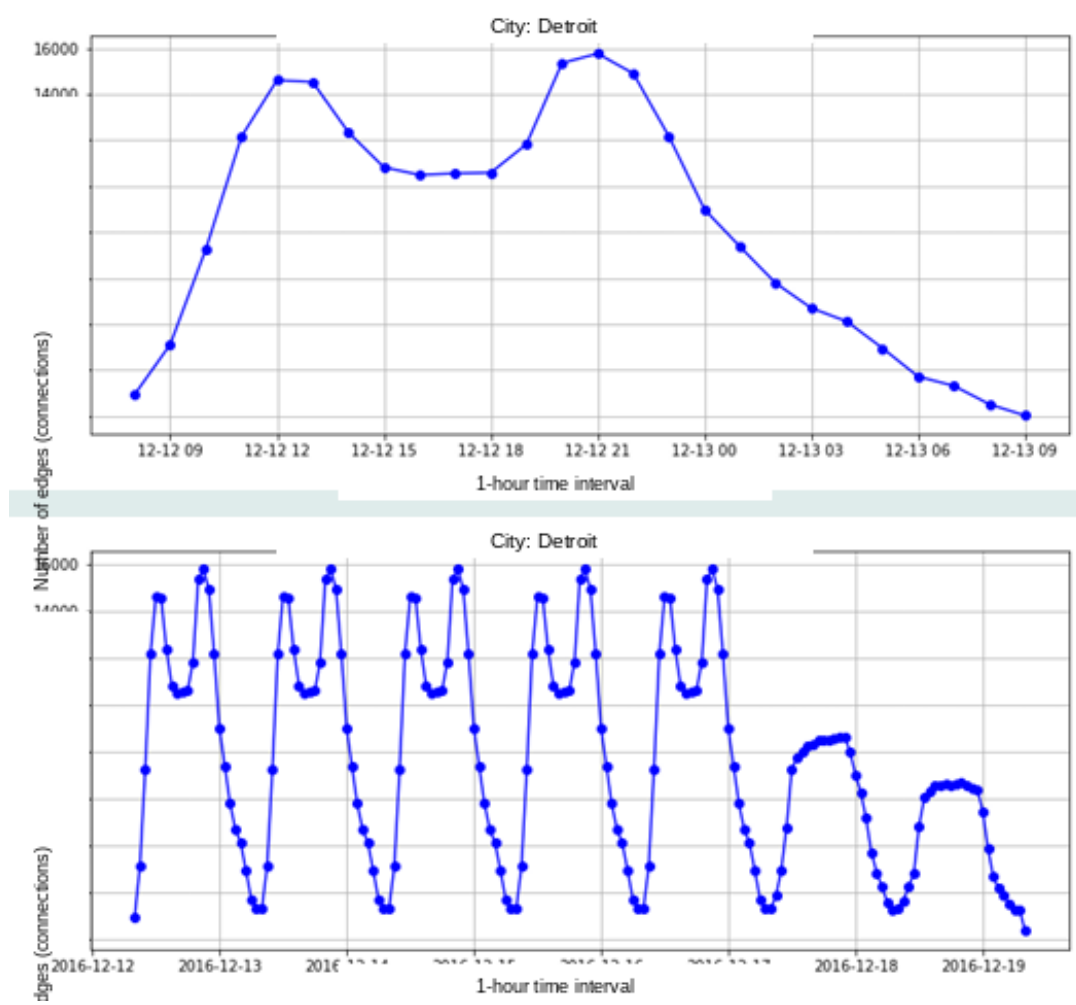


Figure 4: Number of connections by one-hour interval in Detroit: daily file (top) and weekly file (bottom).

7.3 Paris

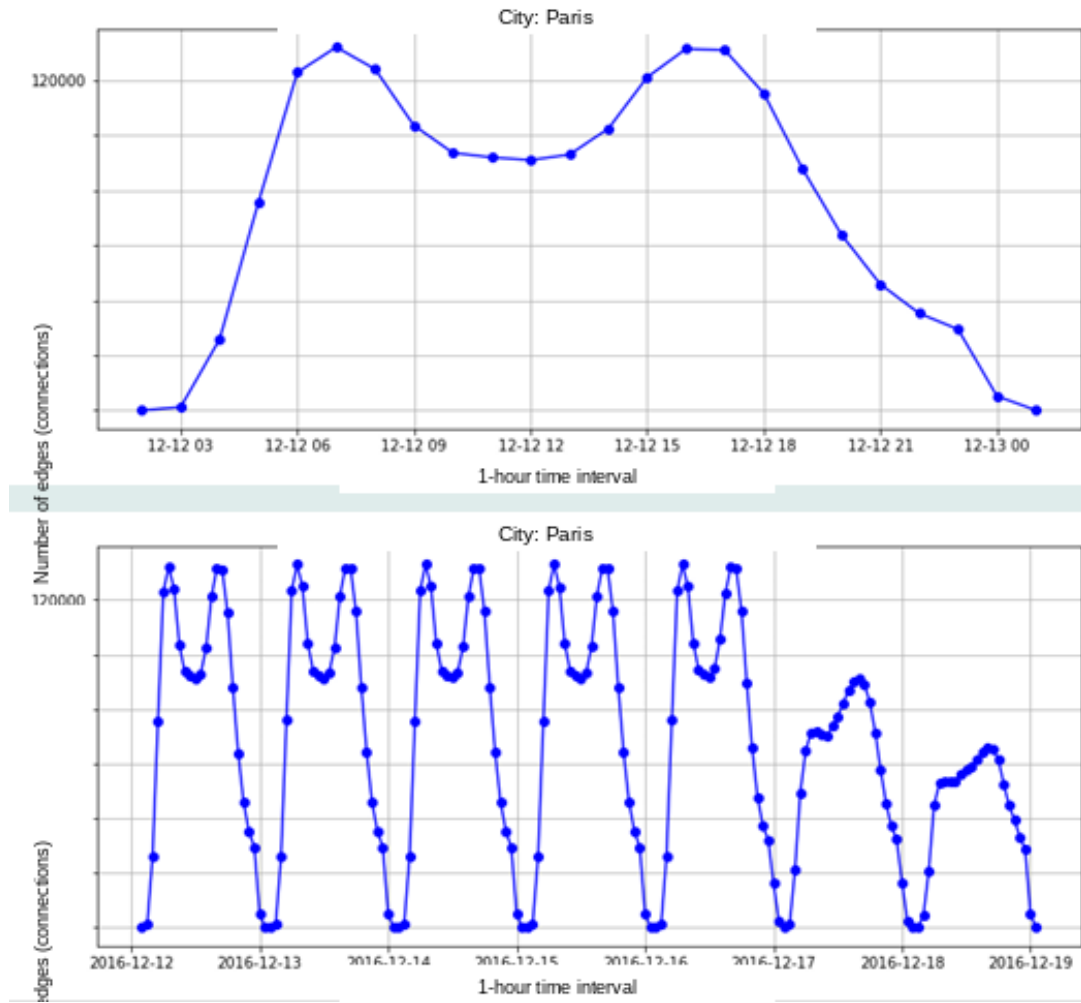


Figure 5: Number of connections by one-hour interval in Paris: daily file (top) and weekly file (bottom).

7.4 Rome

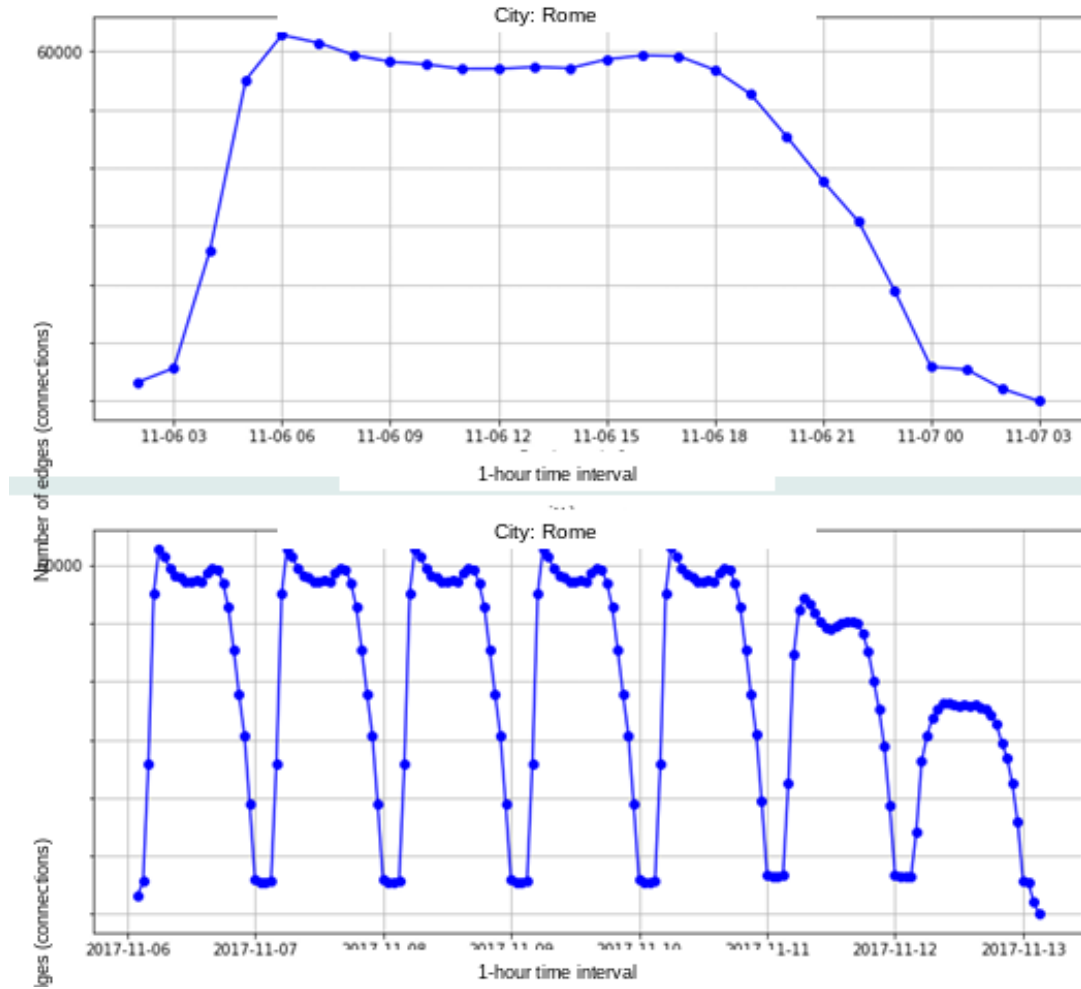


Figure 6: Number of connections by one-hour interval in Rome: daily file (top) and weekly file (bottom).

Across all four cities, the weekly curves reproduce a strong daily cycle. Activity rises rapidly after the low-service overnight period, remains high during daytime hours and falls again in the evening. The final days shown in the weekly files have lower service levels than the preceding weekdays, producing a visible reduction in the number of connections.

7.5 Network Size and Execution Time

The analysis also records execution time as a function of graph size. Larger temporal networks require more processing time, and the relationship is clearly increasing for both the daily and weekly files.

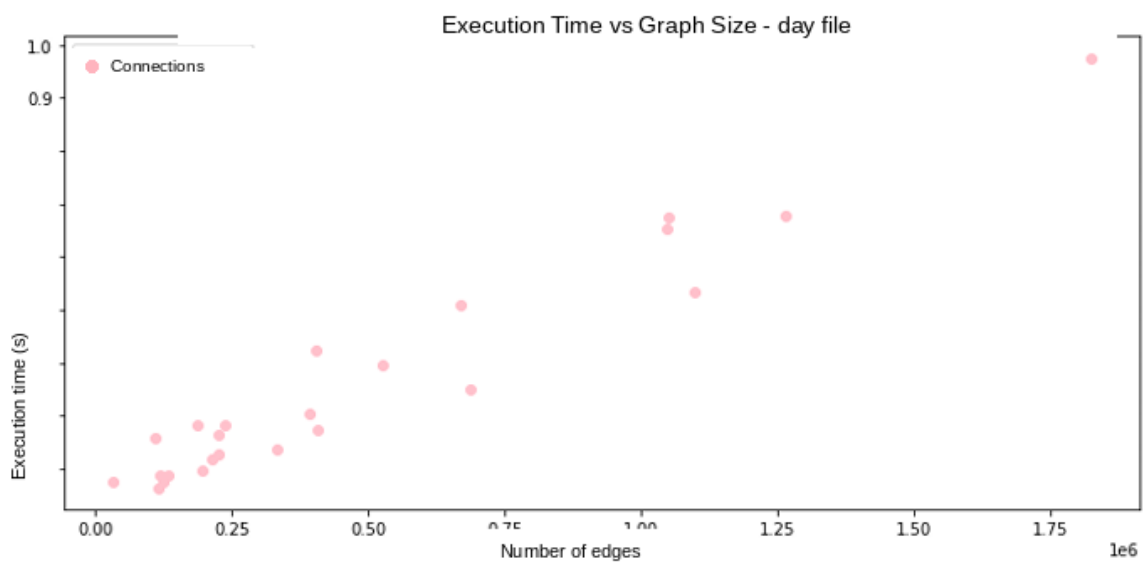


Figure 7: Execution time versus number of edges for the daily files.

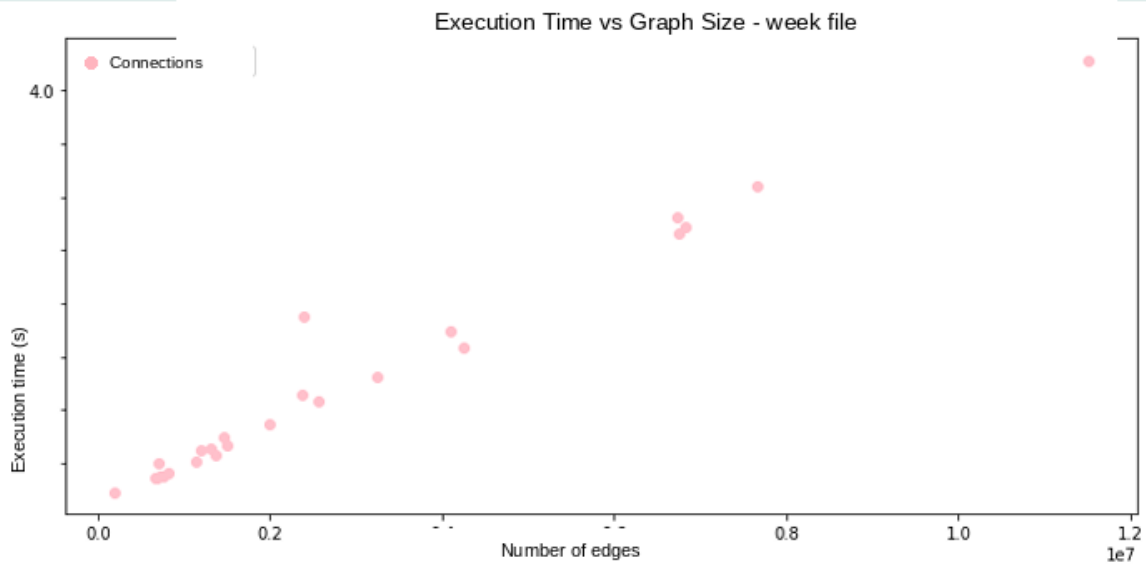


Figure 8: Execution time versus number of edges for the weekly files.

Table 4: Network size and execution-time comparison.

City	Nodes	Day edges	Week edges	Day time (s)	Week time (s)	Edge ratio
Canberra	2764	124305	703614	0.1842	0.5032	5.660
Detroit	5683	214863	1317208	0.2199	0.6328	6.132
Paris	11950	1823872	11507679	0.9759	4.2837	6.311
Rome	7869	1051211	6828376	0.6747	2.7165	6.497

On average, the weekly files contain about 6.15 times as many connections as the daily files, while the execution time increases by a smaller average factor of about 3.51 for this analysis.

8 Task 5: Time-Dependent Reachability

The final task measures accessibility from selected high-out-degree stops. For each starting node, the algorithm determines how many stops can be reached within 30 minutes, 1 hour and 2 hours. The computation uses an *earliest arrival time* algorithm for temporal networks. Connections are processed in chronological order, and a connection can be used only if its departure occurs after the earliest known arrival at its origin and its arrival remains within the allowed time horizon.

This approach is related to shortest-path reasoning, but the edge ordering and waiting times make the problem explicitly temporal. The result is not simply the number of topologically nearby stops; it measures which stops are realistically reachable according to the timetable.

Table 5: Reachable stops from selected Canberra nodes in the daily dataset.

Starting node	30 min	1 h	2 h
Civic Interchange Bay 6, Alinga St (14026)	13	39	115
London Cct Legislative Assembly (1486)	22	325	1749
Morisset St After Crawford St (32495)	45	75	87
Canberra Centre City (1249)	103	474	2154
Soward Way Homeworld (57)	186	519	2192
Comrie St opp Erindale Centre Plt 2 (161)	75	462	2163
Launceston St Phillip Oval (651)	85	336	1619
Brierly St Cooleman Court Plt 2 (1103)	100	535	2195
Corinna St Westfield Woden (1160)	135	412	2214
Streeton Drive after Unwin Pl (1196)	57	356	1857

The results show that a high out-degree does not automatically imply the best time-based accessibility. Timetable structure, departure times, transfers and downstream connectivity all influence how many destinations can be reached within a given time window.

9 Computational Performance

The weekly dataset is substantially larger than the daily dataset, so most algorithms require more time. File loading shows the largest absolute cost, increasing from approximately 5 minutes 58 seconds for the daily data to 44 minutes 48 seconds for the weekly data. Task 1 and Task 2 also scale strongly with the larger edge set, while the diameter computation grows more moderately.

Table 6: Execution times reported for the main procedures.

Procedure	Day	Week	Week/Day
File loading	5 min 58.2212 s	44 min 47.6525 s	7.5027
Task 1	18.1220 s	2 min 58.6406 s	9.857
Task 2	0.1439 s	0.9925 s	6.898
Task 3	16.2017 s	23.1551 s	1.429
Task 5	2 min 51.8669 s	12 min 16.2981 s	4.285

10 Conclusions

The project demonstrates how graph algorithms can be combined with timetable information to study public transport networks from complementary perspectives. Static graph measures identify

structurally important stops and describe the overall topology, while temporal analyses reveal service intensity and realistic accessibility.

The comparison between `day.csv` and `week.csv` also provides a useful computational perspective: the weekly networks contain roughly six times as many connections, but the increase in execution time depends strongly on the algorithm. Finally, the earliest-arrival analysis shows why temporal information is essential for transport applications: connectivity alone does not determine how efficiently passengers can move through a network.